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In-situ surface inspection for wire-arc directed energy deposition integrating 3D topography reconstruction, defect detection and roughness measurement

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ABSTRACT

In wire-arc directed energy deposition (WA-DED), the in-situ surface inspection is imperative for acquiring 3D topographies, detecting weld bead surface defects, and measuring surface roughness to facilitate autonomous quality evaluation and repair path generation. However, existing offline inspection methods fail to provide comprehensive, real-time surface characterization. To address this limitation, a laser vision-based in-situ inspection system, where 3D topography reconstruction, defect detection and roughness measurement are innovatively integrated into a unified real-time scanning task, is developed. The inspection system consists of three subprocesses running simultaneously. By employing the novel laser centerline extraction algorithm CIICEA alongside the novel spatial curve fitting algorithm K-QCBFA, the complete 3D topographies of the printed geometries are reconstructed with a high precision of 0.247 mm. Surface hump and depression defects on the weld bead are identified and classified with a high accuracy of 97.44 % by applying the specific defect detection criterion derived from the analysis of the weld bead height difference curve. Utilizing the surface datum plane determined by the RANSAC algorithm, the geometrical indices for surface roughness, including R_z, R_a, S_z and S_a, are quantitatively evaluated with errors of $\leq$ 0.1558 mm. Experimental validations demonstrate that the system provides comprehensive multi-source surface information during the WA-DED process, capabilities not achievable with single-function inspection technologies, thus enabling sufficient data for effective part quality improvement.

1. Introduction

Compared to Laser Powder Bed Fusion (LPBF), Electron Beam Deposition (EBD) and Laser Directed Energy Deposition (LDED), wire-arc directed energy deposition (WA-DED) exhibits significantly greater potential for manufacturing and repairing larger and more complex engineering parts (Fig. 1) due to its lower equipment cost, higher deposition rate, larger bead width and greater layer thickness [1,2]. However, the intricate and rapid thermal cycles, coupled with dynamic and variable deposition parameters and trajectories, result in geometries printed by WA-DED with poor forming quality and low dimensional accuracy. This issue has long been intractable and remains unavoidable, as illustrated in Fig. 1.

3D topographies, defects and roughness serve as essential indices for evaluating the surface quality of deposited components. Uneven 3D topographies, numerous surface defects, and high surface roughness could seriously degrade the performance of components (e.g., strength, wear resistance and fatigue life), thus diminishing the advantages of WA-DED in industrial production [7–9]. To ensure the quality of components during the WA-DED process, weld bead surface defects and surface roughness are leveraged as inputs into negative feedback mechanisms for regulating various deposition parameters (welding speed, wire feed speed, overlap distance, welding current, welding trajectory, etc.) [10,11], while being utilized to provide timely alerts regarding abnormalities to prevent accumulated errors from leading to component rejections [12]. To obtain high-precision components after deposition, 3D topographic data of the surfaces is used to automatically generate milling repair paths and deposition repair paths [13].

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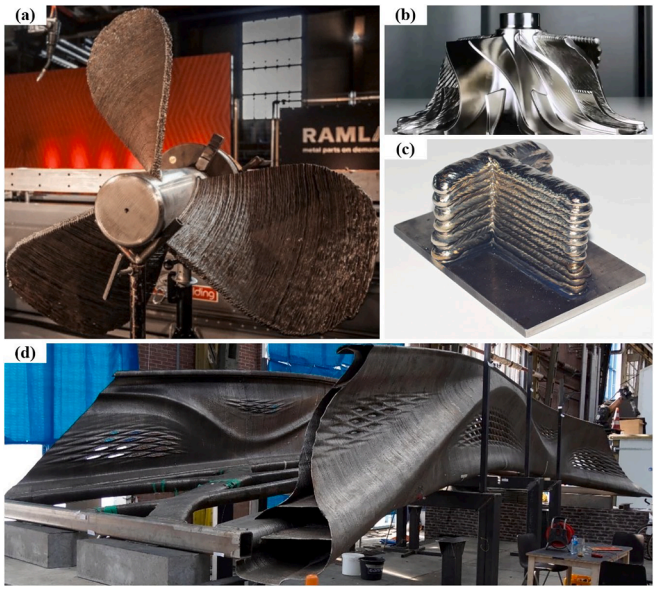

Fig. 1. Engineering parts printed by WA-DED. (a) Propeller [3]. (b) Stainless steel impeller [4]. (c) Rib [5]. (d) Steel bridge [6].

Consequently, developing an efficient in-situ surface inspection method is a crucial step toward advancing WA-DED into a more mature and sophisticated industrial technology [14,15].

Liu et al. [16] reconstructed 3D topographies of 3D printed parts using spatially correlated pixels obtained via binocular stereo vision. Similarly, Li et al. [17] utilized the binocular stereoscopic vision to monitor the 3D surface topography of the powder bed and fusion area. However, the algorithmic complexity for solving spatial stereo matching relationships between overlapping pixels within binocular vision is extremely high. Additionally, binocular vision techniques suffer from larger depth measurement errors. Given these limitations, the laser vision, being capable of perceiving 3D spatial information, are employed for reconstructing 3D topographies of workpieces [13,18–22]. Under actual deposition conditions, factors such as reflective lights, ambient lights and metal fumes introduce defects in laser stripes, including discontinuities, jump points, voids and burrs. Traditional point cloud reconstruction algorithms often struggle to preserve local geometrical surface details and frequently fail to reconstruct high-precision and complete 3D topographies from countless defective laser stripes. Therefore, developing a reliable strategy for reconstructing the surface geometrical morphology remains an open challenge.

Factors such as thermal deformation, fluctuating deposition trajectory, gravitational force, unstable gas blowing force, complex bubble force and rotary electromagnetic force make it difficult to control the solidification state of the liquid metal with strong fluidity, resulting in humps or depressions on the weld bead surface. Solutions have been presented for defect detection for additive manufacturing (AM) have been presented [10,23–32]. Those solutions may be classified as either indirect, where CMOS/CCD cameras [10,23], infrared (IR) thermography sensors [24–26], optical pyrometers [27], electrical signal sensors [28], or other externally assisted equipment are utilized to in-situ monitor the arc, molten pool, local thermal field, temperature, current or voltage to predict defects; and direct methods, where the component itself rather than critical process features is monitored by CMOS/CCD cameras [29,30] and laser vision sensors [31,32]. Compared to indirect prediction methods, direct detection techniques employing 2D or 3D vision provide highly intuitive visualization of defect locations and dimensions. Notably, laser vision-direct detection approaches, including the one presented here, demonstrate significant potential for 3D surface quality assessment. Kaji et al. [18] developed a deep-learning-based in-situ surface anomaly detection methodology for LDED-PF. The deep neural network called RandLA-Net was carried out to process the 3D point cloud data obtained by laser scanner, and the overall average accuracy of 91.3 % and mean IoU of 83.3 % were achieved. Chen et al. [20] inputted 3D surface point cloud data of LAAM-printed components to the KNN, achieving a defect identification accuracy of 93.15 %. Nevertheless, none of these studies is related with WA-DED. In surface defect detection studies for WA-DED, Tang et al. [13] mapped the captured 3D point cloud to a 2D height topography image, and detected depression regions and salient regions by image segmentation and contour extraction. Unlike [13], Huang et al. [19] employed the support vector machine (SVM) to identify bulge and collapsing defects and small pore defects in the 2D height topography image. Obviously, the algorithms required to convert the 3D point cloud into the 2D height image are complex and prone to obscuring information about minor defects. Thus, Liu et al. [21] directly analyzed the 2D curvature of the weld bead height curve and the 3D curvature of the weld bead point cloud, offering important information about the location, type, and volume of pores, humps and bulges on the weld bead for effective interlayer repair. Indeed, the computational load of the algorithms for analyzing the 2D curvature of the weld bead height curve and the 3D curvature of the weld bead point cloud increases significantly as the measured surface area expands. Moreover, interface reflections and tip effects severely compromise curvature calculation accuracy.

The characteristics of the layer-by-layer deposition process dictate that the component's surface deviates from the ideal smooth and flat state, exhibiting inherent higher roughness [33]. Moreover, surface defects could also further exacerbate the component's surface roughness. There are two types of surface roughness determination methods: prediction [11,22,34–39] and measurement. For prediction methods, complex multi-physics deposition phenomena (layer thickness, temperature, printing angle, printing speed, etc.) are correlated with surface roughness. For instance, the regression analysis algorithm was applied to model the relationship between the surface roughness and process parameters such as layer height, wall thickness, nozzle temperature, and printing speed [35]. Furthermore, a variety of deep learning models, including CFNN [11], GA-ANFIS [22], ML [36], GPR [37], DNN [38] and Random Forest [39], are also used in prediction approaches. In contrast, measurement methods offer a more intuitive quantitative assessment for surface roughness with the assistance of 3D vision. For instance, Xiong et al. [40] proposed a laser vision-based methodology for quantitatively calculating the surface roughness of parts deposited by WAAM. Similarly, Yu et al. [41] utilized the laser vision to measure the surface roughness of parts deposited by WAAM and subsequently applied DTC algorithm to evaluate the weld quality. Although the studies [40,41] provided the online roughness measurement methodologies using laser vision, the roughness indices assessed were notably limited. Additionally, no systematic analysis was performed to explore the intrinsic relationships between various roughness indices and surface quality.

Based on the above analysis, the following critical research gaps and corresponding requirements for surface inspection of wire-arc directed energy deposition can be summarized: Current research lacks comprehensive in-situ surface inspection strategies for WA-DED that integrate 3D topography reconstruction, defect detection and roughness measurement; To address this gap, there is a critical need to design the point cloud reconstruction algorithm capable of accurately capturing the complete geometrical topography of the surface without distorting local features; Simultaneously, the development of the straightforward and efficient method for detecting weld bead surface defects remains an urgent priority; Furthermore, establishing robust multi-index roughness assessment strategies that clarify intrinsic relationship between roughness parameters and surface quality is critical for advancing systematic WA-DED quality evaluation.

A novel in-situ surface inspection system for WA-DED based on laser vision is developed in this paper. 3D topography reconstruction, defect detection and roughness measurement are innovatively integrated into a

unified laser scanning task. The novel laser centerline extraction algorithm CIICEA and the novel spatial curve fitting algorithm K-QCBFA are designed to reconstruct the complete 3D topographies of the printed geometries. Surface defects are identified and classified by analyzing the weld bead height difference curve in conjunction with the established defect detection criterion. Utilizing the obtained 3D topography data, the surface datum plane is determined by the RANSAC algorithm, then various roughness indices are measured according to the geometrical relationship between each index and the datum plane, and finally the physical relationships between these indices and the surface quality are analyzed. On the one hand, this system could provide critical decision-making information for quality control during the WA-DED process. The information includes defect categories, defect locations, defect sizes, surface roughness parameters (R_z, R_a, S_z S_a), which provide a reliable foundation for deciding whether to continue deposition or repair the component before further deposition. On the other hand, this system could also provide essential supporting data for the subsequent surface treatments of the component. The data include the position of the datum plane, coordinates of the highest point, coordinates of the lowest point, and maximum height difference, which are used to automatically identify milling and deposition repair areas, and plan repair paths for both processes. Compared with existing inspection methods [19–22,40,41], the inspection strategy designed in this paper could offer a more comprehensive evaluation for the surface quality of components during WA-DED process, thereby advancing the level of automation in WA-DED production.

2. Experimental system

2.1. System setup

Fig. 2 presents the in-situ surface inspection system for WA-DED, which consists of a KUKA 6-axis welding robot, a shielding gas, a wire feeder, a welder, a laser vision sensor (Hangzhou Hikvision Robotics Co., Ltd.), and an industrial PC. The parameters of the laser vision sensor used are shown in Table 1. The laser vision sensor is fixed at the end of the robot and moves with the welding gun. In-situ surface inspection is performed during welding intervals. The robot achieves the switching between welding and inspection tasks through pose transformation. The welding direction during deposition remains aligned with the scanning direction during inspection. In the case of in-situ surface inspection, the robot is programmed to execute a precise pose transformation to project the line laser onto the side surface of the deposited component and subsequently scan it along the welding direction. Throughout the scanning process, the emission window of the laser line maintains a distance of 240 mm from the surface to be inspected, with the laser line oriented perpendicular to the scanning direction. The laser vision sensor communicates with the industrial PC in real time via the TCP/IP protocol. Corresponding programs in the industrial PC process and analyze in real time the large number of 2D laser stripe images captured by the laser vision sensor, thus realizing in-situ surface inspection for WA-DED.

Table 1
Parameters of laser vision sensor.

Model	Laser wavelength	Optimum scanning height	Optimum measure length	Line width
MV-DP2240-01*H*	405 nm	240 mm	140 mm	0.35 mm

2.2. System calibration

Once the system is constructed, the spatial relationship between the sensor frame and the robot frame is established, as illustrated in Fig. 3.

Determining the transformation between the vision coordinate system (O_C-$X_CY_CZ_C$) and the tool coordinate system (O_H-$X_HY_HZ_H$) is the first step for the in-situ surface inspection. The procedure of finding the geometrical relationship between the sensor frame and the robot frame is known as hand-eye calibration [42].

From [43], the transformation logic between the sensor frame and robot frame can be described as

$$\begin{bmatrix} P_B \\ 1 \end{bmatrix} = \begin{bmatrix} R_B & T_B \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_H & T_H \\ 0 & 1 \end{bmatrix} \begin{bmatrix} P_C \\ 1 \end{bmatrix} \tag{1}$$

where P_B are the 3D coordinates of spatial points in the robot base coordinate system (O_B-$X_BY_BZ_B$), P_C are the 3D coordinates of spatial points in the O_C-$X_CY_CZ_C$, $\begin{bmatrix} R_B & T_B \\ 0 & 1 \end{bmatrix}$ are the transformation matrix between the

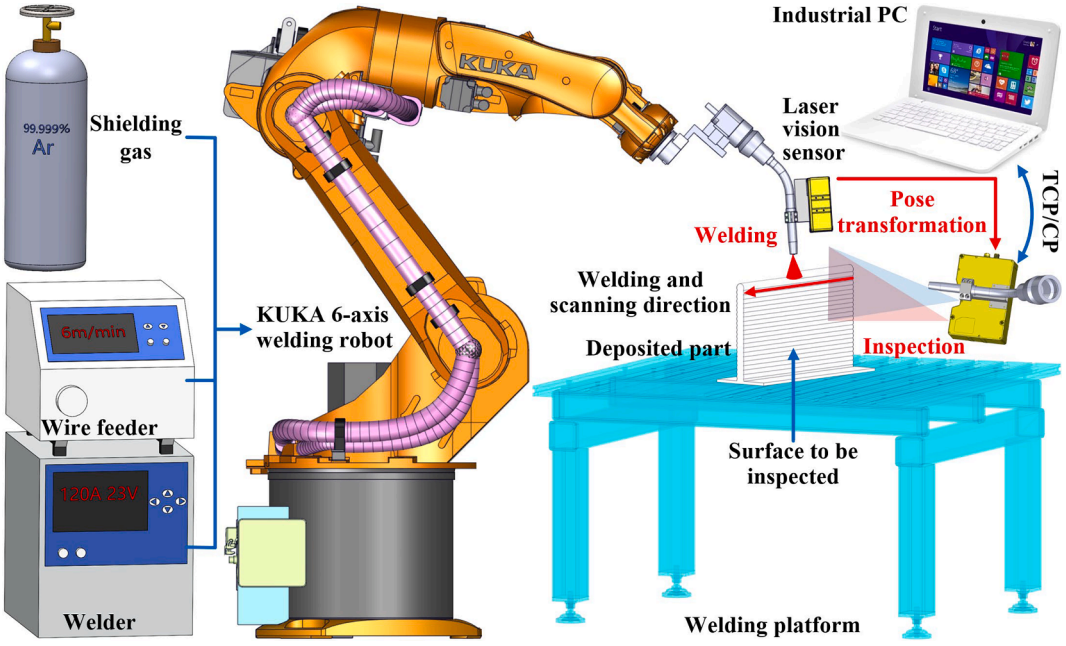

Fig. 2. In-situ surface inspection system for WA-DED.

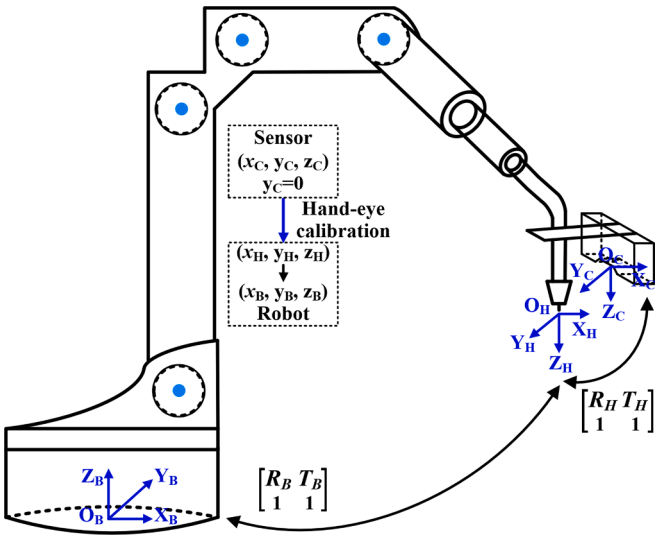

Fig. 3. Coordinate frames in the in-situ surface inspection system.

O_H-$X_HY_HZ_H$ and the O_B-$X_BY_BZ_B$, $\begin{bmatrix} R_H & T_H \\ 0 & 1 \end{bmatrix}$ is are hand-eye relationship between the O_C-$X_CY_CZ_C$ and the O_H-$X_HY_HZ_H$. Calibration aims at calculating the sensor's rotation matrix $R_H \in \mathscr{R}^3$ and the sensor's translation vector $T_H \in \mathscr{R}^3$ under the condition that the known robot's rotation matrix $R_B \in \mathscr{R}^3$ and the known robot's translation vector $T_B \in \mathscr{R}^3$. A particular high-precision geometrical sphere with a 30 mm radius and G10 sphericity as per ISO 3290 standard is used for hand-eye calibration. The welding platform is precision-machined with an array of regularly spaced through-holes, each with a 15 mm radius. For calibration, the calibration sphere is precisely positioned above the designated through-hole. Through the combined action of the sphere's own weight and the mechanical constraint imposed by the through-hole geometry, the calibration sphere is firmly embedded and stationary on the welding platform, as shown in Fig. 4. The geometrical sphere is subjected to translational scans and rotary scans. Point clouds on the surface of the sphere containing the radius information and the position information of sphere's center are instrumental in calibrating the rotation matrix and translation vector in the hand-eye relationship.

According to Eq. (1), points on the surface of the sphere can be described as:

$$P_{B_S}^{[i]} = R_B^{[i]} R_H P_{C_S}^{[i]} + R_B^{[i]} T_H + T_B^{[i]} \tag{2}$$

where $i = 1,2,3\cdots n$, $P_{B_S}^{[i]}$ are the 3D coordinates of the sphere's center in the O_B-$X_BY_BZ_B$, $P_{C_S}^{[i]}$ are the 3D coordinates of the sphere's center in the O_C-$X_CY_CZ_C$. The expressions of $P_{C_S}^{[i]}$ are as follows:

$$P_{C_S}^{[i]} : \begin{cases} x_{C_S}^{[i]} = x_{C_C}^{[i]} + \vec{a}\,t \\ y_{C_S}^{[i]} = y_{C_C}^{[i]} + \vec{b}\,t \\ z_{C_S}^{[i]} = z_{C_C}^{[i]} + \vec{c}\,t \end{cases} \tag{3}$$

where ($x_{C_C}^{[i]}$, $y_{C_C}^{[i]}$, $z_{C_C}^{[i]}$) are the 3D coordinates of the center of tangent circle of the sphere in the O_C-$X_CY_CZ_C$. Laser point clouds projected onto the surface of the sphere exhibit circular stripe characteristics, and these circular point clouds encapsulate information regarding the tangent circle of the sphere. By employing least squares fitting on these circular point clouds, the radius r of the tangent circle and the coordinates ($x_{C_C}^{[i]}$, $y_{C_C}^{[i]}$, $z_{C_C}^{[i]}$) of the center of the circle in the O_C-$X_CY_CZ_C$ could be obtained. The expressions of t are as follows:

$$t = \frac{\sqrt{R^2 - r^2}}{\sqrt{a^2 + b^2 + c^2}} \tag{4}$$

where$R = 30$ mm, ($\vec{a}$, $\vec{b}$, $\vec{c}$) are the normal vectors of the laser plane.

Under translational scans with the R_B constant and the T_B gradually varying, rotation matrix R_H is calibrated.

The sphere is stationary in the robot frame, i.e., $P_{B_S}^{[1]} = P_{B_S}^{[i]}$, then:

$$R_B R_H P_{C_S}^{[1]} + R_B T_H + T_B^{[1]} = R_B R_H P_{C_S}^{[i]} + R_B T_H + T_B^{[i]} \tag{5}$$

Eq. (5) can also be rewritten as:

$$R_B R_H (P_{C_S}^{[i]} - P_{C_S}^{[1]}) = T_B^{[i]} - T_B^{[1]} \tag{6}$$

$$R_H (P_{C_S}^{[i]} - P_{C_S}^{[1]}) = R_B^T (T_B^{[i]} - T_B^{[1]}) \tag{7}$$

Eq. (7) can be abbreviated as:

$$AR_H = B \tag{8}$$

Based on the 7 translational scans, Eq. (8) is expanded as

$$\begin{bmatrix} A_{[1]} \\ A_{[2]} \\ \vdots \\ A_{[6]} \end{bmatrix}_{6\times3} \bullet R_{H3\times3} = \begin{bmatrix} B_{[1]} \\ B_{[2]} \\ \vdots \\ B_{[6]} \end{bmatrix}_{6\times3} \tag{9}$$

Eq. (9) is called overdetermined equation group and is solved by Singular Value Decomposition (SVD).

$$\begin{bmatrix} A_{[1]} \\ A_{[2]} \\ \vdots \\ A_{[6]} \end{bmatrix}_{6\times3} = U^1{}_{6\times6} \begin{bmatrix} \sum_{3\times3} \\ 0 \end{bmatrix} V^{1\,T}{}_{3\times3} \tag{10}$$

By Eq. (9) and Eq. (10), the R_H can be calculated:

$$R_H = V^1 \Sigma^{-1} U^1(:, 1:3)^{\mathrm{T}} B \tag{11}$$

Under rotary scans with the R_B gradually varying and the T_B constant, translation vector T_H is calibrated.

Eq. (5) can also be rewritten as:

$$R_B^{[1]} R_H P_{C_S}^{[1]} + R_B^{[1]} T_H = R_B^{[j]} R_H P_{C_S}^{[j]} + R_B^{[j]} T_H \tag{12}$$

$$(R_B^{[j]} - R_B^{[1]}) T_H = R_B^{[1]} R_H P_{C_S}^{[1]} - R_B^{[j]} R_H P_{C_S}^{[j]} \tag{13}$$

Eq. (13) can be abbreviated as:

$$CT_H = D \tag{14}$$

Based on the 7 rotary scans, Eq. (14) is expanded as:

$$\begin{bmatrix} C_{[1]} \\ C_{[2]} \\ \vdots \\ C_{[6]} \end{bmatrix}_{6\times3} \bullet T_{H3\times3} = \begin{bmatrix} D_{[1]} \\ D_{[2]} \\ \vdots \\ D_{[6]} \end{bmatrix}_{6\times3}$$

Eq. (14) is also called overdetermined equation group and is solved by Singular Value Decomposition (SVD).

$$\begin{bmatrix} C_{[1]} \\ C_{[2]} \\ \vdots \\ C_{[6]} \end{bmatrix}_{6\times3} = U^2{}_{6\times6} \begin{bmatrix} \sum_{3\times3} \\ 0 \end{bmatrix} V^{2\,T}{}_{3\times3} \tag{15}$$

By Eq. (14) and Eq. (15), the T_H can be calculated:

$$T_H = V^2 \Sigma^{-1} U^2(:, 1:3)^{\mathrm{T}} D \tag{16}$$

According to Eq. (11) and Eq. (16), the hand-eye relationship $\begin{bmatrix} R_H & T_H \\ 0 & 1 \end{bmatrix}$ is obtain.

To evaluate the accuracy of hand-eye calibration, multi-directional and multi-angle sphere radius measurement experiments were

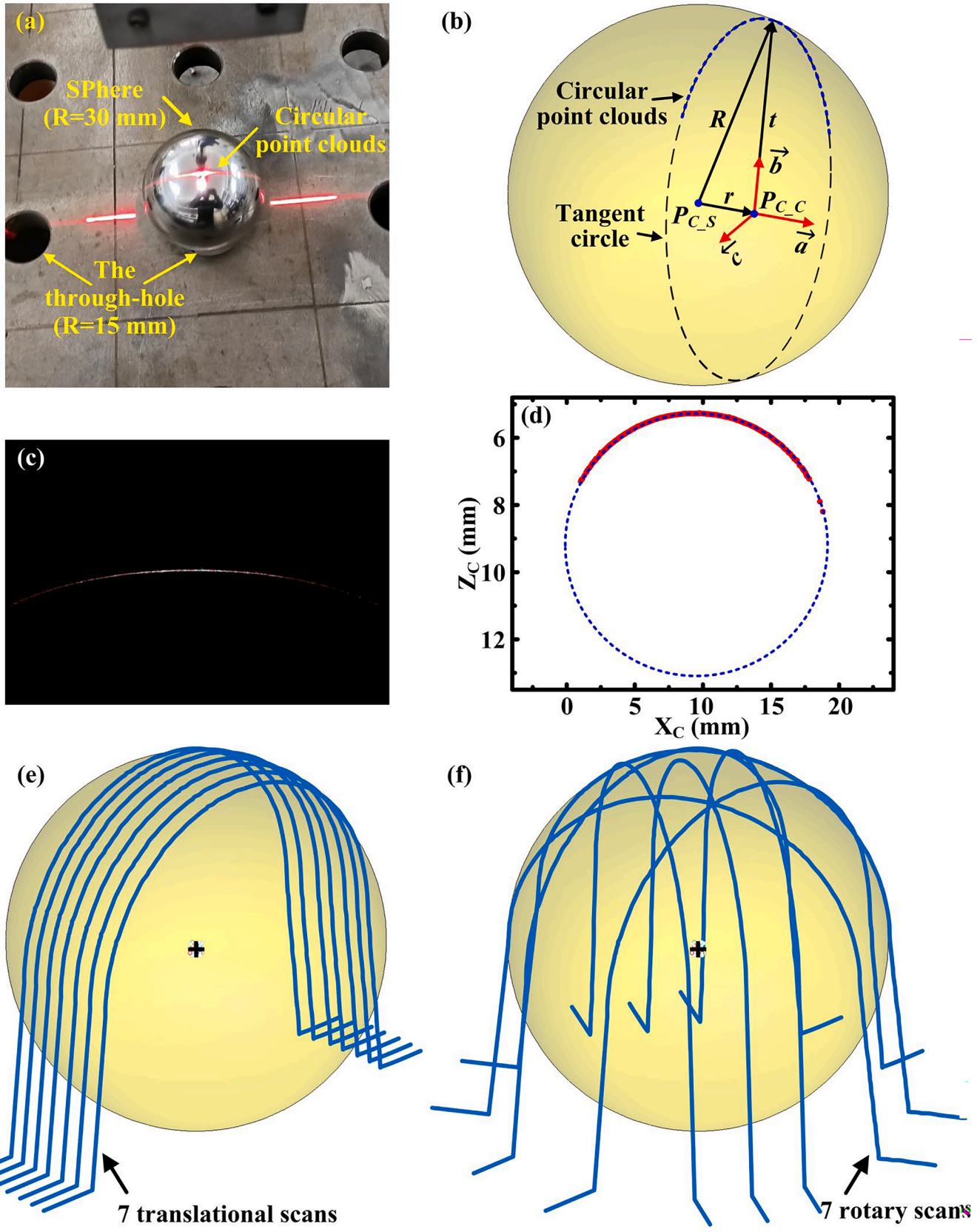

Fig. 4. Hand-eye calibration diagram. (a) Geometrical sphere. (b) Calculation of spherical center. (c) Spherical surface laser image. (d) Point clouds on spherical surface. (e) Calibrating rotation matrix R_H. (f) Calibrating translation vector T_H.

Table 2
Calibration errors.

Measurement pose of robot	ΔR(mm)
Pose 1	−0.1375
Pose 2	0.0656
Pose 3	−0.0435
Pose 4	0.1029
Pose 5	0.1423
Average absolute error	0.0984

conducted. Based on the hand-eye relationship determined by Eq. (11) and Eq. (16), the 3D point clouds of spherical surfaces acquired in the vision coordinate system were transformed into the robot base coordinate system. These point clouds reconstruct the 3D structure of the sphere and contain the sphere radius information. The point cloud reconstruction error, i.e. the calibration error, is quantified by the sphere radius measurement error. The acquired sphere radius errors at five different measurement poses are presented in Table 2. The results show an average absolute calibration error of 0.0984 mm, demonstrating sufficient accuracy for in-situ surface inspection requirements in wire-arc directed energy deposition applications.

2.3. Surface quality inspection strategy

The designed in-situ surface inspection strategy consists of five modules: data acquisition, data processing, 3D topography reconstruction, defect detection and roughness measurement, as illustrated in Fig. 5. The data processing module comprises three steps: centerline extraction, spatial curve fitting, and 3D point cloud transformation. From Fig. 5, the laser stripe images collected from the data acquisition module inevitably contain interferences such as background noise, reflection lights and burrs, which reduce the purity of the stripe images and obscure their effective information. To extract the stripe centerline from strong interference, a Context Information-based Iterative Centerline Extraction Algorithm (CIICEA) is innovatively designed. Discontinuities, jump points, voids and other defects at local positions such as corners and peaks destroy the integrity of the laser stripe. Moreover, the limited number of pixels in the laser stripe is insufficient to meet the density requirements for reconstructing 3D topography. To achieve accurate and complete high-density laser point clouds, a K-means-based quasi-uniform cubic B-spline fitting algorithm (K-QCBFA) is innovatively developed. The interpolated and fitted point cloud is bound in the 2D pixel space, which cannot accurately represent the geometrical characteristics of the surface, and therefore needs to undergo 3D coordinate transformation using Eq. (1). By systematically arranging and combining numerous 3D laser stripe point clouds in the robot base coordinate system according to the order of acquisition in real time, 3D topography reconstruction can be effectively achieved. In the defect detection module, the weld bead height curve is initially determined by extracting the geometrical point cloud at the top of the 3D topography. Subsequently, a specific defect detection criterion is formulated by analyzing the height difference characteristics of the weld bead height curve. Finally, humps and depressions are accurately identified and classified using the criterion. The surface datum plane, which is the most crucial prerequisite for surface roughness measurement, is derived through iterative computations using the RANSAC algorithm. Concurrently, the quantitative assessment of the surface roughness is conducted by calculating various roughness indices.

3. Data processing

3.1. Centerline extraction

There are two types of methods for extracting the centerline of the

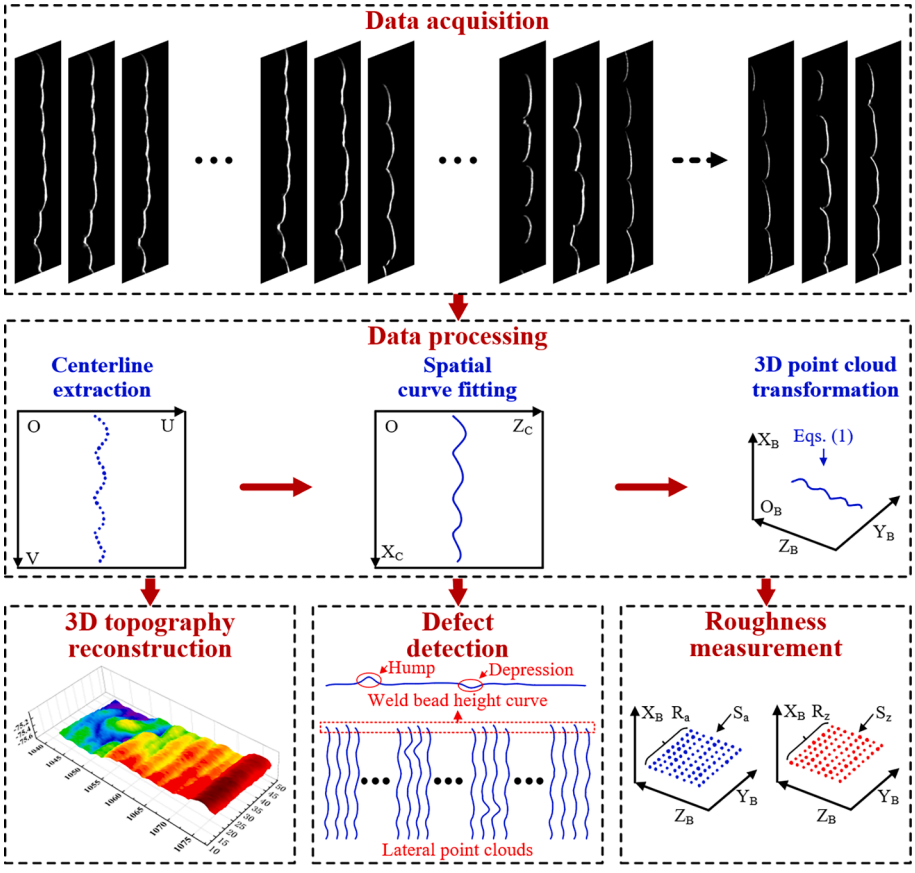

Fig. 5. In-situ surface inspection strategy.

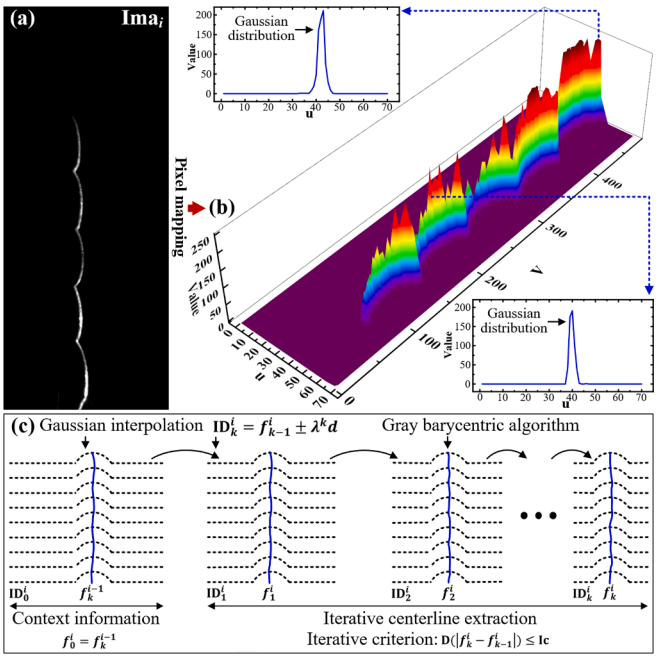

Fig. 6. Context information-based iterative centerline extraction algorithm. (a) Acquired original images. (b) Pixel domain of interest. (c) Principle of CIICEA.

laser stripe [44]: geometry-based methods and energy-based methods. Although geometry-based methods, including the skeleton thinning algorithm and the geometrical center algorithm, can rapidly and efficiently extract the centerline of the laser stripe, their extraction accuracy is limited to the pixel level. In contrast, energy-based methods, such as the gray barycentric algorithm [45], the Steger algorithm [46] and the Hough algorithm [47], are capable of achieving sub-pixel accuracy.

It should be emphasized that these traditional methods are particularly vulnerable to interference from background noise, reflective lights, and burrs. Under complex and changing working conditions, such as the in-situ surface inspection scenario in this paper, traditional centerline extraction methods struggle to achieve high adaptability and success rates while meeting the requirements for sub-pixel level extraction precision. Therefore, a Context Information-based Iterative Centerline Extraction Algorithm (CIICEA) is innovatively proposed, as depicted in Fig. 6. The CIICEA extracts the stripe centerline through the following steps:

Step 1: extract the pixel domain of interest (PDI) $\mathrm{ID}^i_{\mathrm{PDI}}$ from the laser stripe image Img_i according to Eq. (17).

$$\mathrm{ID}^i_{\mathrm{PDI}} = \{(\mathrm{u},\mathrm{v})|\mathrm{Img}_i(\mathrm{u},\mathrm{v}) > 0\} \tag{17}$$

where $\mathrm{Img}_i(\mathrm{u},\mathrm{v})$ represents the gray value of the pixel point (u,v). $\mathrm{ID}^i_{\mathrm{PDI}}$ is the set of pixel points with non-zero gray values within the image Img_i.

Step 2: extract the pixel domain ID^i_0 by performing Gaussian interpolation (Eq. (18)) on pixels in each column within $\mathrm{ID}^i_{\mathrm{PDI}}$. Results of Gaussian interpolation are shown in Fig. 7, demonstrating that most of original pixel points are interspersed by interpolated pixel point curves. These curves closely adhere to the distribution trend of original pixel points, strongly confirming the validity and accuracy of Gaussian interpolation algorithm proposed in this paper.

$$f(x_g) = \sum_{j=1}^{n} \omega_j \bullet f(x_j) \tag{18}$$

where ω_j and $f(x_j)$ are the weights and gray values of known points x_j, respectively. x_g are the points to be interpolated, which are determined according to the equal spacing principle (Eq. (19)).

$$x_g = \min(x_j) + (g-1) \bullet \frac{\max(x_j) - \min(x_j)}{num - 1},\ g = 1 \cdots num \tag{19}$$

where *num* represents the total number of points generated by the interpolation.

The expression of ω_j is as follows:

$$\omega_j = \frac{G(x_g - x_j)}{\sum_{j=1}^{n} G(x_g - x_j)} \tag{20}$$

where the expression of Gaussian function $G(x_g - x_j)$ is as follows:

$$G(x_g - x_j) = \exp\left(-\frac{(x_g - x_j)^2}{2\sigma^2}\right) \tag{21}$$

where σ stands for the standard deviation of the Gaussian function. The parameter σ serves as a critical factor in Gaussian interpolation, directly governing the smoothness and detail retention of the interpolation result. A larger σ tends to make the interpolated curve smoother, suitable for data with lower peaks and wider distributions. Conversely, Smaller σ is more likely to retain local details and is suitable for data with higher peaks and narrower distributions. From Fig. 6(b), the curves of each column of pixels in $\mathrm{ID}^i_{\mathrm{PDI}}$ demonstrate characteristics of higher peaks and narrower distributions, hence a smaller σ should be chosen. Notably, to prevent overfitting while maintaining adequate detail resolution, σ should not be reduced excessively. Through empirical validation, this study establishes an optimal σ value of 0.35. As shown in Fig. 7(b), the interpolated point cloud curve not only accurately retains

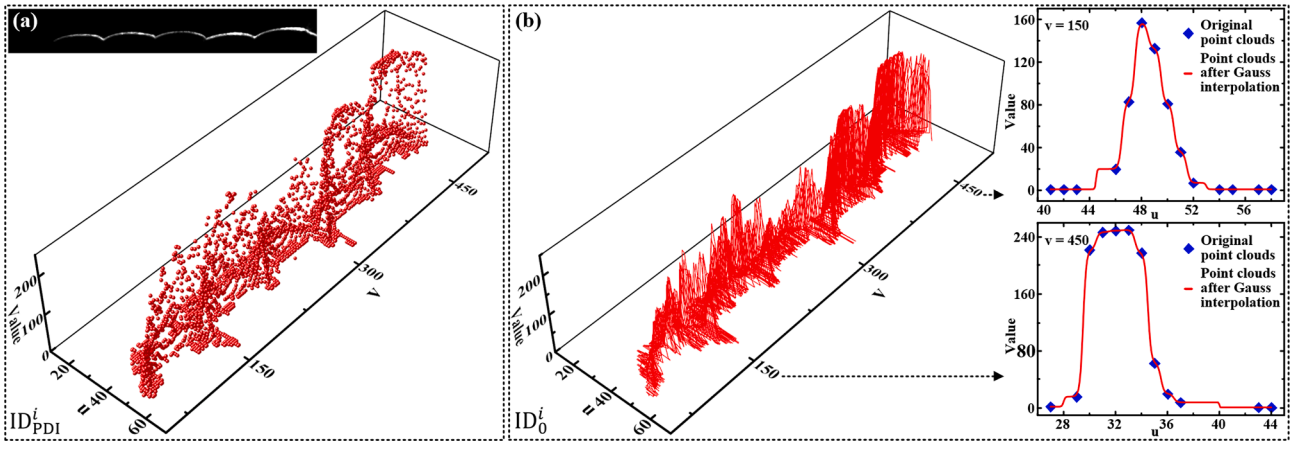

Fig. 7. Results of Gauss interpolation. (a) Pixel point clouds in PDI. (b) Pixel point clouds after Gauss interpolation.

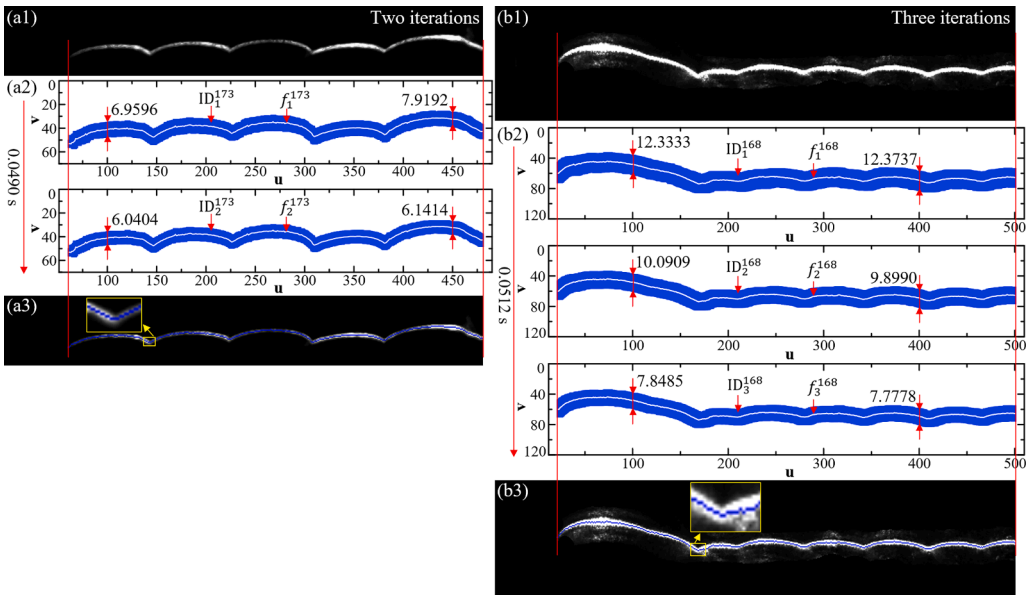

Fig. 8. Results of extracting centerlines using CIICEA. (a1) Original images under normal interferences. (b1) Original images under strong interferences. (a2-b2) The process of CIICEA extracting centerlines. (a3-b3) Mapping relationship of laser stripes and extracted centerlines.

the variation trend of the original data, but also traverses the original point cloud with remarkable precision.

Step 3: extract the pixel domain $\mathrm{ID}_k^i(k=1,2\cdots)$ using iterative operation (Eq. (22)) based on the obtained centerline f_{k-1}^i.

The principle of extracting ID_k^i is as follows:

$$\mathrm{ID}_k^i : \begin{cases} f_{k-1}^i - \lambda^k d \\ f_{k-1}^i + \lambda^k d \end{cases} \tag{22}$$

where λ and d are the iteration factor and iteration parameter, respectively, with$\lambda = 0.8$ and$d = 8$. f_{k-1}^i is obtained from ID_{k-1}^i using the gray barycentric algorithm. Additionally, according to the context information, $f_0^i = f_k^{i-1}$.

Step 4: set the iterative criterion (Eq. (23)). When the spatial relationship between f_k^i and f_{k-1}^i conforms to Eq. (23), the iterative operation stops, otherwise, the iterative operation continues.

$$\mathrm{D}\left(\left|f_k^i - f_{k-1}^i\right|\right) < \mathrm{Ic} \tag{23}$$

where D denotes the variance function, and Ic is the iteration termination criterion. $\mathrm{D}\left(\left|f_k^i - f_{k-1}^i\right|\right)$ is used to calculate the spatial geometrical difference between the centerline f_k^i acquired in the k-th iteration and the centerline f_{k-1}^i acquired in the (k-1)-th iteration. The iterative process terminates when this geometrical difference falls within Ic pixels. To extract the centerline within the sub-pixel level, i.e., the geometrical difference between f_k^i and f_{k-1}^i is less than one pixel, the Ic should be set to a value less than 1. To preventing infinite loops caused by excessively small Ic values, this study empirically sets Ic to 0.5 through systematic optimization, establishing an optimal balance between computational precision and algorithmic convergence.

Results of extracting centerlines using CIICEA are shown in Fig. 8 and Fig. 9. From Fig. 8 and Fig. 9(a), without resorting to any image preprocessing methods (e.g. gray transform, threshold segmentation, morphological processing, edge detection, etc.), CIICEA is still capable of extracting laser stripe centerlines with high efficiency and accuracy. After each iterative, pixel domains of interest are continuously compressed, and image interferences are progressively removed, which fundamentally explains CIICEA's robustness against dynamic and complex noises (e.g. light spots, spatters, stripe burrs, reflections, etc.), as confirmed by the comparative experiments shown in Fig. 9. Fig. 9(b) and Fig. 9(c) illustrate the stripe centerline extraction results using the gray barycentric algorithm and the skeleton extraction algorithm, respectively. The centerlines extracted by the gray barycentric algorithm are severely deviated from the stripe center. the skeleton extraction algorithm fails to accurately extract the centerline at the stripe end due to pixel compression and refinement operations. Additionally, the algorithm erroneously extracts noise skeletons, which is detrimental to accuracy. In contrast, even under strong interferences, the centerlines extracted by CIICEA remain precisely centered within the stripes. Moreover, The CIICEA processes each image frame in approximately 50 ms, achieving a centerline extraction frame rate of 20 fps, satisfying real-time in-situ inspection requirements. The CIICEA algorithm can be summarized in **Algorithm 1**.

Algorithm 1. CIICEA
Input: The laser stripe image stream Img_i
The centerline f_k^{i-1}
Output: The centerline f_k^i
Begin
  Extract the pixel domain of interest $\mathrm{ID}_{\mathrm{PDI}}^i$
  Obtain ID_0^i by Gaussian interpolation for pixels in each column within $\mathrm{ID}_{\mathrm{PDI}}^i$
  Set $f_0^i = f_k^{i-1}$, λ=0.8, d=8, Ic=0.5, k=1
  for u ← 1 to size (ID_1^i) step 1 do
    $\mathrm{ID}_1^i = f_0^i \pm \lambda^1 d$
  endfor
  Extract the centerline f_1^i within ID_1^i by gray barycentric method
  while $\mathrm{D}\left(\left|f_k^i - f_{k-1}^i\right|\right) \geq \mathrm{Ic}$
    $k = k + 1$
    for e ← 1 to size (ID_{k-1}^i) step 1 do
      $\mathrm{ID}_k^i = f_{k-1}^i \pm \lambda^k d$
    endfor
    Extract the centerline f_k^i within ID_k^i by gray barycentric method
  endwhile
End

3.2. Spatial curve fitting

The analysis for the centerline point cloud extracted by CIICEA indicates that: First, the number of discrete pixel points in the centerline is

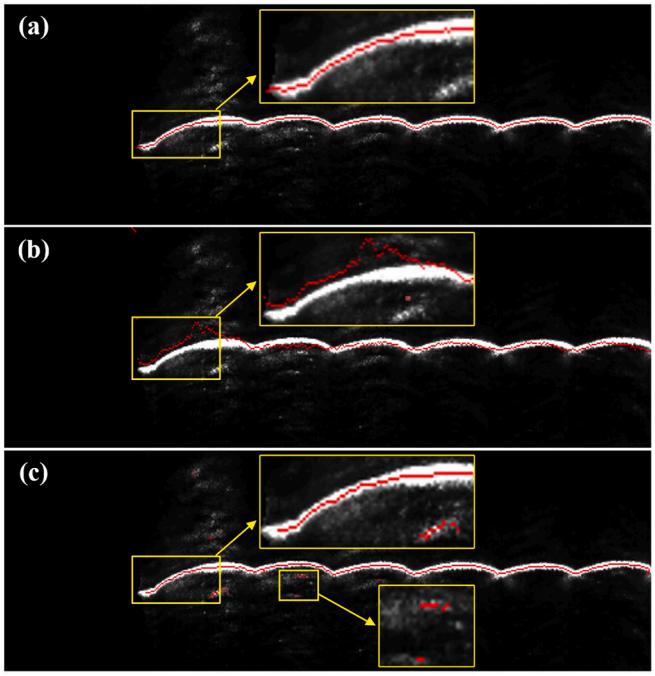

Fig. 9. The comparative experiments for extracting centerlines. (a) Centerlines extracted by CIICEA. (b) Centerlines extracted by the gray barycentric algorithm. (c) Centerlines extracted by the skeleton extraction algorithm.

limited by the camera's resolution constraints. The point cloud density of the centerline is insufficient to satisfy the requirements for in-situ quality inspection. Second, factors such as interface reflections, metal fumes and lens contamination could induce defects in the centerline, including discontinuities, jump points and voids. The damaged centerline is also insufficient to satisfy the requirements for in-situ quality inspection.

For accurately reconstructing the complete geometrical morphology of the surface, interpolation and fitting for the centerline point cloud acquired by CIICEA are essential. Common fitting methods include polynomial fitting [21], Bezier curve fitting and B-spline curve fitting. Polynomial fitting exhibits higher fitting errors when applied to complex point clouds (e.g., wavy laser stripe point clouds). Bezier curve fitting necessitates the manual selection of control points, resulting in lower efficiency. B-spline fitting is prone to overfitting when applied to non-uniformly distributed point clouds (e.g., point clouds at discontinuities). Therefore, a K-means-based quasi-uniform cubic B-spline fitting algorithm (K-QCBFA) is innovatively designed, as shown in Fig. 10. The K-QCBFA fits the stripe centerline through the following steps:

Step 1: divide the stripe center point cold $\mathrm{p}_n(n=1,2,\cdots)$ into K clusters $\mathrm{p}^K_{n_K}(K, n_K=1,2,\cdots)$ using unsupervised machine learning K-means algorithm (Eq. (24)).

$$\mathrm{ED}=\arg\min\sum_{i=1}^{K}\left(\sum_{x\in S_i}\|x-\mu_i\|^2\right) \tag{24}$$

where ED stands for Euclidean distance, K is the number of clusters, μ_i is the centroid of cluster S_i, and S_i represents the i-th cluster partitioned from p_n. The centroid μ_i is calculated as:

$$\mu_i=\frac{1}{|S_i|}\sum_{x\in S_i}x \tag{25}$$

The K-means algorithm iteratively updates S_i to minimize ED until convergence, yielding the final clusters $\mathrm{p}^K_{n_K}$.

Step 2: perform quasi-uniform cubic B-spline fitting (Eq. (26)) on the K clusters $\mathrm{p}^K_{n_K}$.

$$F^K(t^K)=[p_0^K,p_1^K,\cdots,p_{n_K}^K]\begin{bmatrix}B^K_{0,e}(t^K)\\ B^K_{1,e}(t^K)\\ \vdots\\ B^K_{n_K,e}(t^K)\end{bmatrix}=\sum_{i=0}^{n_K}p_i^K\cdot B^K_{i,e}(t^K) \tag{26}$$

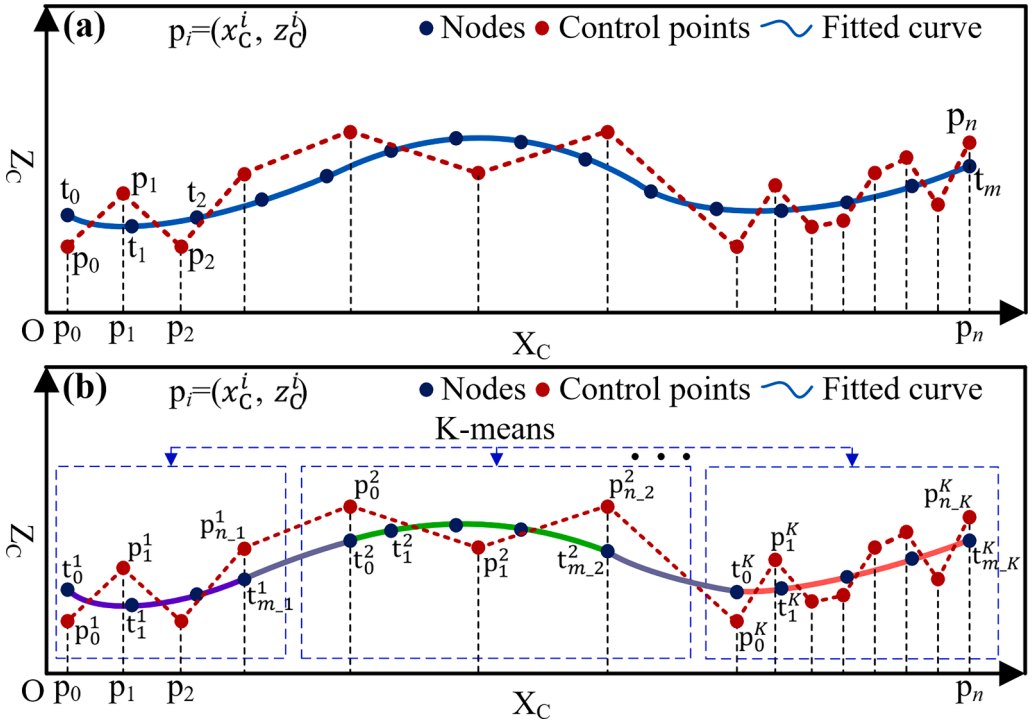

Fig. 10. K-means-based quasi-uniform cubic B-spline fitting algorithm. (a) Traditional quasi-uniform cubic B-spline fitting algorithm. (b) Modified quasi-uniform cubic B-spline fitting algorithm.

where e is the order of the B-spline curve, $B_{i,e}^{K}(t^{K})$ is B-spline basis function, and the expression for $B_{i,e}^{K}(t^{K})$ is as follows:

$$B_{i,1}^{K}(t^{K}) = \begin{cases} 1, t^{K} \in [t_i^K, t_{i+1}^K] \\ 0, else \end{cases}$$
$$B_{i,e}^{K}(t^{K}) = \frac{t^{K} - t_i^K}{t_{i+e-1}^K - t_i^K} B_{i,e-1}^{K}(t^{K}) + \frac{t_{i+e}^K - t^{K}}{t_{i+e}^K - t_{i+1}^K} B_{i+1,e-1}^{K}(t^{K}) \quad (27)$$

where the node vector t^{K} is defined as:

$$t^{K} = \{t_0^K, t_1^K, \cdots, t_i^K, t_{i+1}^K \cdots, t_{m_K}^K\} \quad (28)$$

where t_i^K are nodes, m_K+1 is the number of nodes, and $t_i^K < t_{i+1}^K$.

Step 3: combine $F^{K}(t^{K})$ to achieve the fitted centerline F_{Img_i}.

Results of fitting centerlines using K-QCBFA are shown in Fig. 11. From Fig. 11, the proposed K-QCBFA algorithm can effectively analyze and extract the clustering characteristics of the control points, and adaptively regulate and assign nodes based on these characteristics. This capability of K-QCBFA not only avoids the overfitting of the B-spline curve, but also ensures that the fitted spatial curve accurately matches the spatial distribution of the discrete stripe center point cloud, as illustrated in Fig. 11(c). Fig. 11(a) shows the spatial curves fitted by the polynomial fitting algorithm with significant fitting errors. Fig. 11(b) shows the spatial curves fitted by the quasi-uniform cubic B-spline fitting algorithm with partial overfitting. In contrast, only the spatial curves (Fig. 11(c)) fitted by K-QCBFA are more accurate and reliable. The K-QCBFA algorithm can be summarized in **Algorithm 2**.

Algorithm 2. K-QCBFA
Input: Stripe center point could $p_n(n = 1, 2, \cdots)$
Output: The fitted centerline F_{Img_i}
1. **Begin**
2. Cluster analysis through Eq. (24)
3. Divide $p_n(i = 1, 2, \cdots)$ into K clusters $p_{n_K}^K(K, n_K = 1, 2, \cdots)$
4. **for** u ← 0 to K step 1 do
5. Quasi-uniform cubic B-spline fitting through Eq. (26)
6. Get $F^{K}(t^{K})$ by fitting $p_{n_K}^K$
7. **Endfor**
8. Get F_{Img_i} by combining $F^{K}(t^{K})$
9. **End**

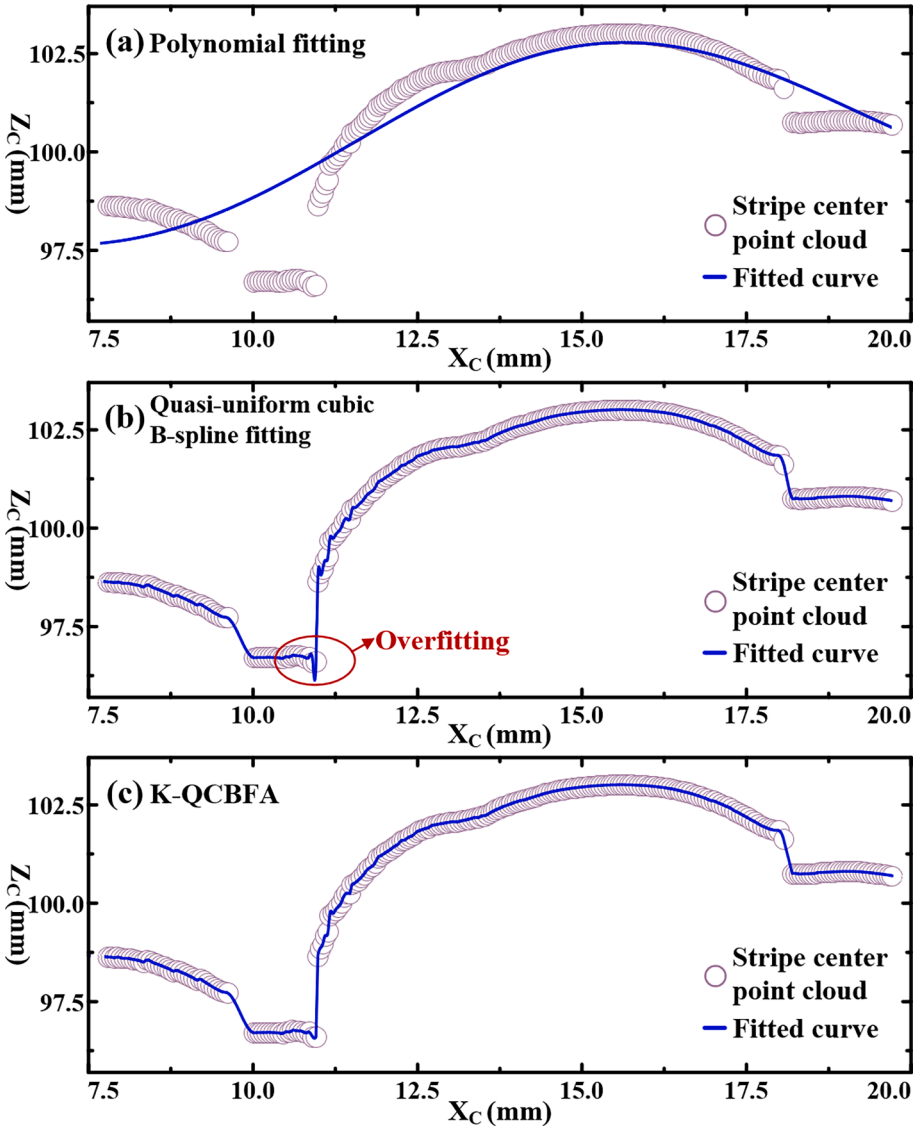

Fig. 11. The comparative experiments for fitting center points. (a) Spatial curves fitted by polynomial fitting algorithm. (b) Spatial curves fitted by quasi-uniform cubic B-spline fitting algorithm. (c) Spatial curves fitted by K-QCBFA.

4. Surface quality inspection objectives

4.1. 3D topography reconstruction

Using Eq. (1), the fitted centerline F_{Img_i} in the vision coordinate system can be transformed into the robot base coordinate system. The geometrical morphology of the surface is reconstructed in real time by systematically arranging and combining numerous 3D centerline point clouds in the robot base coordinate system according to their acquisition sequence. Fig. 12 illustrates a representative 3D topography reconstruction case. The original surface geometrical morphology of the deposited component is shown in Fig. 12(a). The process of in-situ real-time reconstruction is depicted in Fig. 12(b) and Fig. 12(c), with Fig. 12 (c) presenting the complete reconstructed 3D topography. The reconstructed 3D topography, achieved through the laser centerline extraction algorithm CIICEA and the spatial curve fitting algorithm K-QCBFA, effectively preserves local geometrical details while accurately and reliably representing global morphological characteristics, as evidenced in Fig. 12.

4.2. Defect detection

Common weld bead surface defects include humps and depressions, as shown in Fig. 13. These defects disrupt the ideal profile of the weld bead surface. The laser stripes at defect locations display distinct geometrical characteristics, providing a theoretical basis for defect recognition and classification.

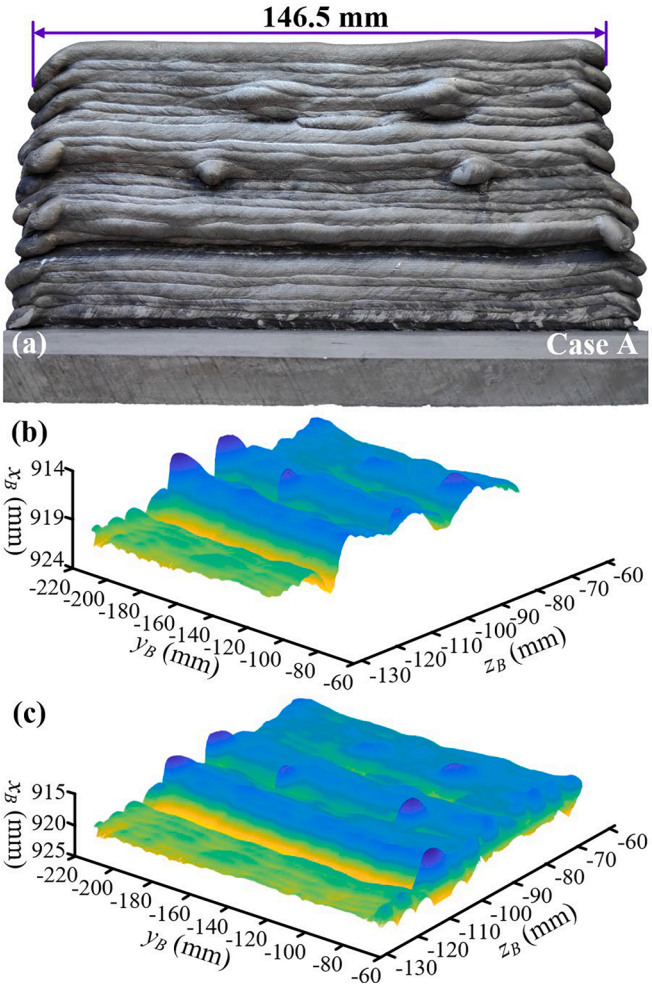

Fig. 12. 3D topography reconstruction process. (a) Case A. (b) Y_B = -213.17 ~ -109.80 mm. (c)Y_B = -213.17 ~ -62.51 mm.

As the most intuitive and accurate index for characterizing the surface quality of the weld bead, the extraction of the weld bead height curve is the primary step in detecting weld bead surface defects. As shown in Fig. 5, the 3D weld bead height curve (Fig. 14) is determined precisely by extracting the geometrical point cloud at the top of the 3D topography. From Fig. 14, the height difference is the essential geometrical characteristic of the weld bead height curve. This characteristic provides critical information about the locations and morphologies of defects, and is thus applied for defect detection. For calculating the height difference accurately, the 3D weld bead height curve must be converted into the 2D weld bead height curve. The 2D weld bead height curve is the projection of the 3D weld bead height curve onto the O_B-Y_BZ_B plane, as illustrated in Fig. 14.

The theoretical weld bead height is determined by analyzing the consistency of 2D weld bead height curve using the RANSAC algorithm. The weld bead height difference curve is defined as the deviation between the measured height curve and the theoretical height curve. The sign (positive or negative) of the height difference values directly indicates the type of defect, while their absolute magnitudes reflect the severity of the defect.

Based on the height differences, a specific defect detection criterion for identifying humps and depressions is established, as shown in Eq. (29).

$$\begin{cases} h_D < cr_D \\ h_H > cr_H \end{cases} \tag{29}$$

where cr_D and cr_H are the depression and hump determination thresholds, respectively, h_D and h_H represent the corresponding height differences. Since the height difference between the measured height and theoretical height at each point on the weld bead directly reflects the severity of defects, the criterion based on the height difference is instituted to detect defects. This criterion essentially defines the tolerance thresholds for defects. The thresholds cr_D and cr_H are selectively configured according to the required quality inspection level. Taking Case A (Fig. 15) as an example, when cr_D is set to 1 mm and cr_H to −1 mm, the detection results indicate 0 hump defects and 2 depression defects. Conversely, when cr_D is reduced to 0.5 mm and cr_H is increased to −0.5 mm, the system identifies 1 hump defect and 2 depression defects. Evidently, larger absolute values of cr_D and cr_H lead to fewer detected defects, whereas smaller absolute values result in increased defect detection sensitivity. In this study, any absolute height difference exceeding 0.5 mm between actual measurements and theoretical values is classified as a defect, corresponding to threshold settings ofcr_D = 0.5 mm andcr_H = -0.5 mm.

The defect detection results for Case A are shown in Fig. 15. Fig. 15 (a) displays the measured height curve alongside the theoretical height curve. Fig. 15(b) illustrates the weld bead height difference curve obtained by subtracting the theoretical height curve from the measured height curve. Fig. 15(c) presents the detection results for the depression. Fig. 15(d) shows the detection results for the hump. 1 hump and 2 depressions were detected in Case A. Fig. 15(e) illustrates the positions of the detected defects, with areas highlighted in blue boxes representing the humps and areas marked in red boxes indicating the depressions.

4.3. Roughness measurement

Surface topography can be broken down into three main components: form, waviness, and roughness. The form refers to the large-scale geometrical characteristics of the surface, reflecting the overall contour or design intent of the workpiece. The waviness describes intermediate-scale periodic undulations on the surface, which exists between form and roughness. The roughness characterizes microscale surface irregularities typically induced by manufacturing processes. In precision machining, precise control of form, waviness, and roughness is essential

Fig. 13. Common weld bead surface defects.

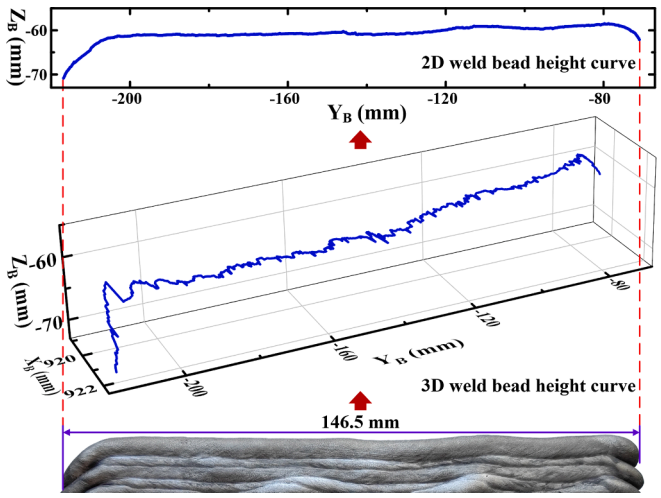

Fig. 14. Weld bead height curve.

to ensure that components meet design requirements. However, for wire-arc directed energy deposition (WA-DED), rigorous separation of these three surface characteristics becomes less critical, primarily because the deposited components will undergo subsequent machining operations (e.g., surface turning) prior to service [13]. In the hybrid manufacturing process combining deposition and turning, rapid and convenient surface quality inspection is a core requirement for process optimization. Characterizing surface quality by directly extracting roughness parameters from surface topographies not only satisfies efficiency requirements but also facilitates quantitative evaluation. Therefore, consistent with the methodologies in references [22,37–41], this study analyzes the acquired 3D surface topographies by directly extracting roughness parameters from these topographies according to their standard definitions, eliminating the need for filtering to separate form, waviness, and roughness.

Evaluation indices for roughness include the maximum height of section contour roughness (R_z), the arithmetic mean deviation of section contour roughness (R_a), the maximum height of surface roughness (S_z), the arithmetic mean deviation of surface roughness (S_a) [48]. R_z and R_a serve as indices for evaluating surface roughness based on local 2D section contour, while S_z and S_a are indices for assessing surface roughness based on global 3D geometrical morphology. The use of these four indices together can provide a very comprehensive evaluation of surface roughness, which accounts for their widespread application in industry [49].

The rules for calculating R_z, R_a, S_z and S_a are shown below:

(1) R_z is defined as the maximum vertical distance from the summit to the valley in the section contour.

$$R_z = \max\{h_convex(x_B^i, z_B^i)\} + \max\{h_concave(x_B^i, z_B^i)\} \tag{30}$$

where $h_convex(x_B^i, z_B^i)$ and $h_concave(x_B^i, z_B^i)$ denote vertical distances from the convex and concave points on the section contour to the datum plane, respectively.

(2) R_a is defined as the arithmetic mean of vertical distances from all points on the section contour to the datum plane.

$$R_a = \frac{1}{N}\sum_{1}^{N} h(x_B^i, z_B^i) \tag{31}$$

where $h(x_B^i, z_B^i)$ denotes the vertical distance from each point on the section contour to the datum plane.

(3) S_z describes the maximum vertical distance from the summit to the valley in the surface.

$$S_z = \max\{h_convex(x_B^i, y_B^i, z_B^i)\} + \max\{h_concave(x_B^i, y_B^i, z_B^i)\} \tag{32}$$

where $h_convex(x_B^i, y_B^i, z_B^i)$ and $h_concave(x_B^i, y_B^i, z_B^i)$ denote vertical distances from the convex and concave points on the surface to the datum plane, respectively.

(4) S_a describes the arithmetic mean of vertical distances from all points on the surface to the datum plane.

$$S_a = \frac{1}{MN}\sum_{j=1}^{M}\sum_{i=1}^{N} h(x_B^i, y_B^i, z_B^i) \tag{33}$$

where $h(x_B^i, y_B^i, z_B^i)$ denotes the vertical distance from each point on the surface to the datum plane.

As previously mentioned, the datum plane is the most crucial prerequisite for evaluating surface roughness. The expression formula for the datum plane, as presented in Eq. (34), is determined using the RANSAC algorithm. The process of determining the datum by the RANSAC algorithm is outlined in **Algorithm 3**.

$$Az_B + By_B + Cx_B + D = 0 \tag{34}$$

For Case A, the determined datum plane and the vertical distance from each point on the surface to the datum plane are illustrated in Fig. 16. The calculated R_z and R_a are presented in Fig. 17. According to Eq. (35), $\overline{R_z}$ and $\overline{R_a}$ are 5.2829 mm and 0.8145 mm, respectively. From Fig. 17(a), R_z is able to judge appearance of the defects on the surface of the deposited component and to locate the positions of the defects. When bulges and concavities appear on the inspection surface, R_z will show a sudden and significant increase at the positions of the defects. From Fig. 17(b), R_a is capable of quantitatively determining the severity of the stair stepping characteristic (the geometrical overlap characteristic of deposited metals). When the stair stepping characteristic at a

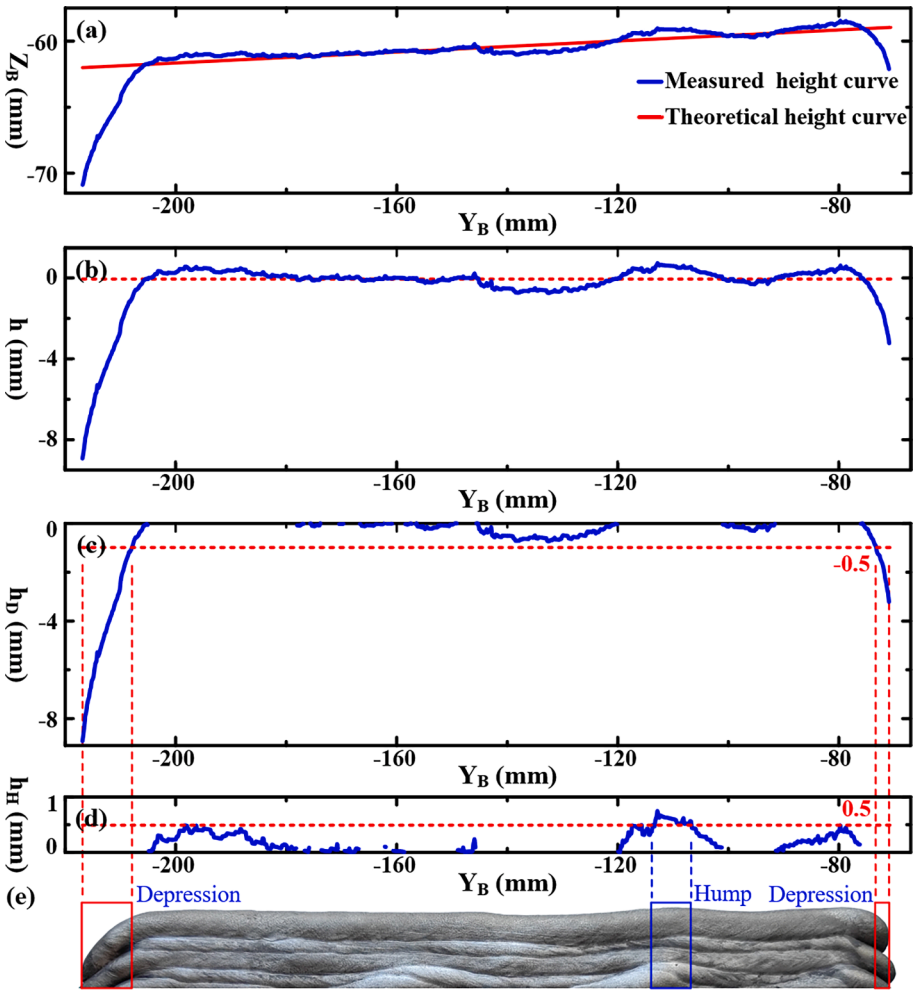

Fig. 15. Defect detection for Case A. (a) Measured height curve and theoretical height curve. (b) Weld bead height difference curve. (c) Detection results for depression. (d) Detection results for hump. (e) Positions of defects.

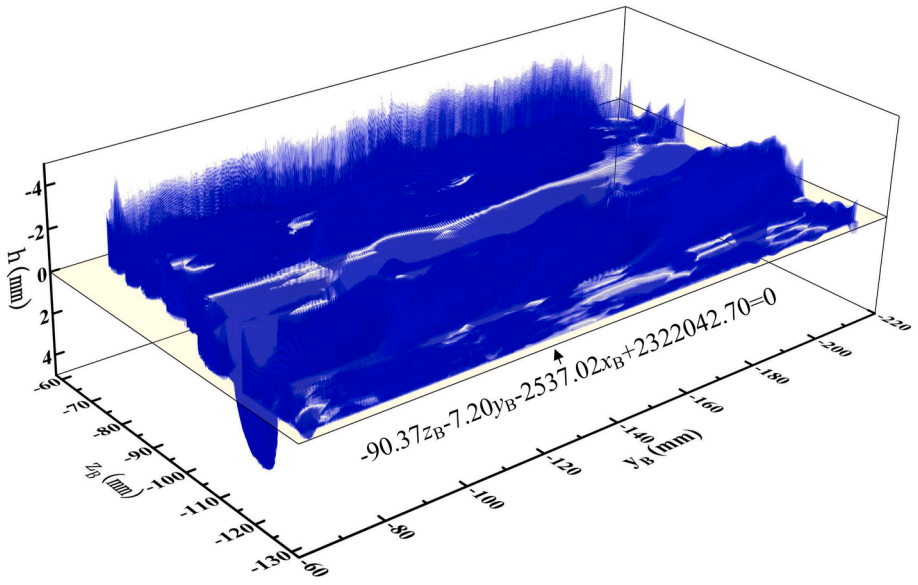

Fig. 16. The datum plane and the vertical distance from each point on the surface to the datum plane.

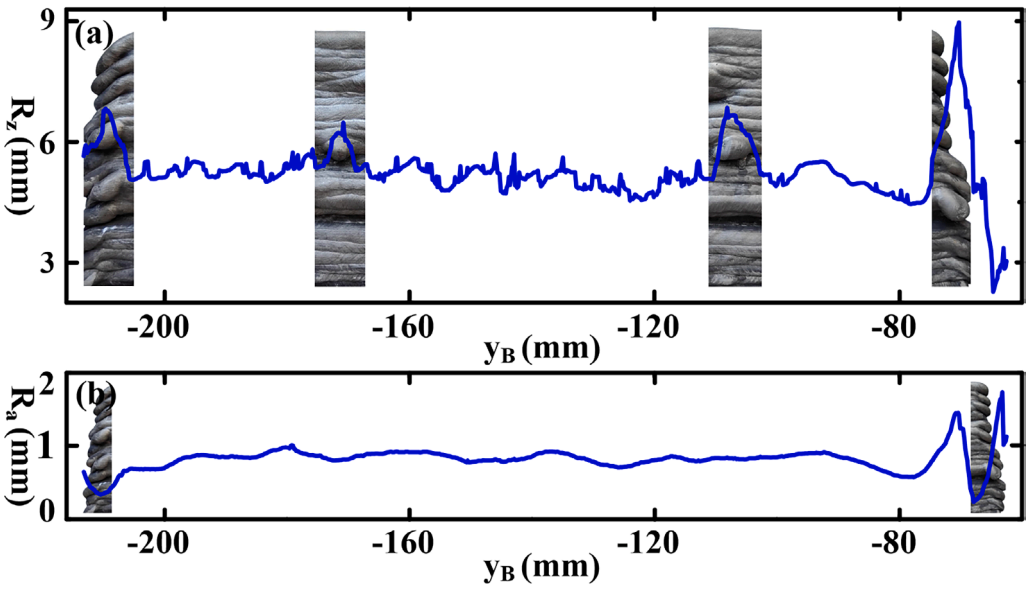

Fig. 17. Section contour roughness. (a) R_z. (b) R_a.

specific location is more pronounced than that in the surrounding areas, R_a will also exhibit a sudden and significant increase at that position. According to Eq. (32) and Eq. (33), S_z=14.0726 mm andS_a = 0.8159 mm.

$$\{\overline{R_z} = \frac{\sum_{\min(y_B)}^{\max(y_B)} R_Z^{y_B}}{\max(y_B) - \min(y_B)} \overline{R_a} = \frac{\sum_{\min(y_B)}^{\max(y_B)} R_a^{y_B}}{\max(y_B) - \min(y_B)} \tag{35}$$

where $R_Z^{y_B}$ and $R_a^{y_B}$ are R_z and R_a on each y_B, respectively.

Algorithm 3. RANSAC plane fitting
Input: The fitted centerline $F_{\mathrm{Img}_i}(i = 1, 2, \cdots)$
Output: The fitted datum plane $\varnothing(x_B, y_B, z_B) = 0$
1. **Begin**
2. Coordinate transformation for F_{Img_i} through Eq. (1)
3. Set the total number of points be N_{totlal}
4. Select three random points to determine initial plane equation
5. Calculate the vertical distance all points to the above plane equation
6. Save the points that meet given vertical distance tolerance
7. Set the number of the saved points be N_{saved}
8. **If** $N_{saved} = N_{totlal}$
9. The above plane equation is $\varnothing(x_B, y_B, z_B) = 0$
10. **else**
11. Determine new plane equation used the saved points
12. **End**
13. Repeat the steps 5–12 for a given number of cycles
14. Set the plane equation obtained last time as $\varnothing(x_B, y_B, z_B) = 0$
15. **End**

5. Case analysis

To comprehensively assess the performance of the proposed in-situ surface inspection system for WA-DED integrating 3D topography reconstruction, defect detection and roughness measurement, two extra single-pass multi-layer depositions (Case B and Case C) are utilized for testing. Fig. 18 illustrates Case B and Case C. The surface properties of Case A, Case B and Case C are presented in Table 3, and these cases with different surface properties facilitate the comprehensive evaluation of the performance of the proposed inspection system. The surface formation states of each case are created through the specific deposition process.

5.1. 3D topography reconstruction results and analysis

The results of 3D topography reconstruction for Case B and Case C

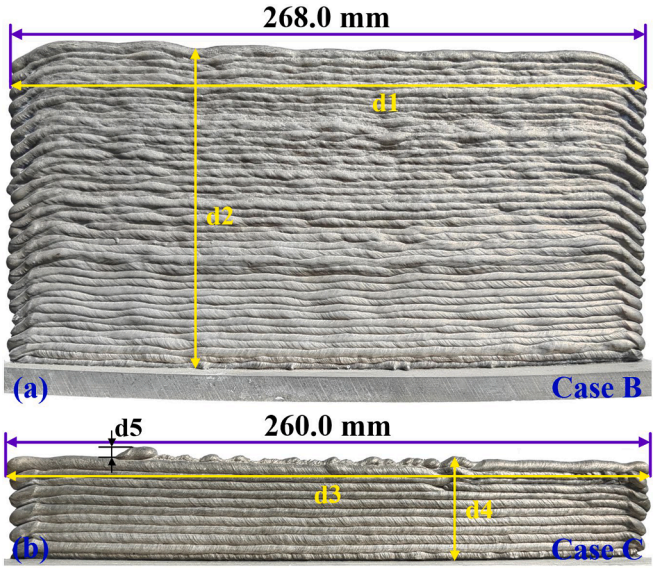

Fig. 18. Extra depositions. (a) Case B. (b) Case C.

Table 3
Surface properties of Case A, Case B and Case C.

Case No.	Weld bead surface defect	Altitude difference	Flatness level
Case A	Small number of defects	Maximum	Minimum
Case B	normal number of defects	Minimum	Medium
Case C	Large number of irregular defects	Medium	Maximum

are depicted in Fig. 19. The reconstructed 3D topographies are devoid of voids, outliers, and other point cloud defects, accurately and effectively reproducing the 3D surface morphologies of the deposited components. In addition, local geometrical details on the surface, such as surface bulges, surface concavities and surface grooves, are also reconstructed realistically and reliably, confirming the absence of geometrical distortions in the reconstruction algorithms.

Based on the reconstructed 3D surface topographies, the determined

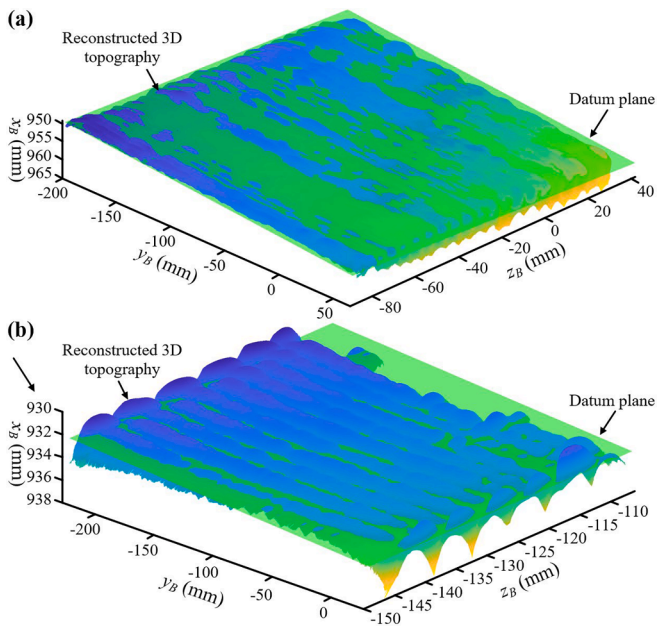

Fig. 19. Results of 3D topography reconstruction. (a) Case B. (b) Case C.

datums for Case B and Case C are $-3.48z_B + 12.57y_B - 1620.86x_B + 1513831.61 = 0$ and $-183.67z_B - 148.65y_B + 5455.20x_B - 5227864.89 = 0$, respectively. Table 4 presents the precision of the 3D topography reconstruction. Compared to manual measurement, the system's reconstruction error is 0.247 mm, satisfying industrial quality inspection accuracy requirements.

5.2. Defect detection results and analysis

Fig. 20(a) and Fig. 21(a) present the measured weld bead height curves and theoretical weld bead height curves for Case B and Case C, respectively. Fig. 20(b) and Fig. 21(b) display the detection results of the depression for Case B and Case C, respectively. Fig. 20(c) and Fig. 21(c) show the detection results of the hump for Case B and Case C, respectively. In Case B, 4 humps and 5 depressions were detected, while in Case C, 12 humps and 14 depressions were detected. Fig. 20(d) and Fig. 21(d) illustrate the positions where the detected defects are located for Case B and Case C, respectively.

The comparison between defect detection results through the algorithm and those through manual inspection is shown in Table 5. In all cases, manual inspection identified 17 humps and 21 depressions, while the algorithm detected 16 humps and 22 depressions. The algorithm mistakenly identified a normal region as a depression and incorrectly identified a hump as a normal region.

Drawing on the evaluation methods for object detection performance of neural network [18,21,29], indices including accuracy, precision, recall and F1 score are used to evaluate the performance of the defect detection algorithm in this paper. The calculation principles of these indices are presented in Eq. (36):

Table 4
Reconstruction errors.

Target No.	Algorithm measurement (mm)	Manual measurement (mm)	Error (mm)
d1	268.175	268.0	0.175
d2	130.416	131.0	0.584
d3	260.107	260.0	0.107
d4	42.020	42.3	0.280
d5	4.512	4.6	0.088
Average error	0.247		

$$\begin{cases} Accuracy = \dfrac{TP + TN}{TP + TN + FP + FN} \\ Precision = \dfrac{TP}{TP + FP} \\ Recall = \dfrac{TP}{TP + FN} \\ F1 = 2 \times \dfrac{Precision*Recall}{Precision + Recall} \end{cases} \tag{36}$$

where:*TP*(Ture Positive): the number of positive samples predicted by the model as the positive class.*FP*(False Positive): the number of negative samples predicted by the model as the positive class.*TN*(Ture Negative): the number of negative samples predicted by the model as the negative class.*FN*(Ture Negative): the number of positive samples predicted by the model as the negative class.

Table 6 gives the calculation results of the evaluation indices. The results demonstrate that the proposed defect detection algorithm is capable of identifying the majority of defects and accurately classifying them, thereby validating the algorithm's effectiveness in defect detection. Notably, the algorithm misidentifies humps and depressions as normal areas with probabilities of 5.88 % and 0 %, respectively, while misidentifying normal areas as humps and depressions with probabilities of 0 % and 4.55 %, respectively. Specifically, the criterion for defect detection is established based on the global geometrical characteristics of the weld bead, while neglecting the local geometrical information, which diminishes the detection accuracy.

5.3. Roughness measurement results and analysis

Fig. 22(a) and Fig. 23(a) present the measurement results of R_z for Case B and Case C, respectively. Fig. 22(b) and Fig. 23(b) display the measurement results of R_a for Case B and Case C, respectively. $\overline{R_z}$ and $\overline{R_a}$ for Case B are 3.9560 mm and 0.4312 mm, respectively, while for Case C, they are 3.0509 mm and 0.2747 mm, respectively. From Fig. 17, Fig. 22 and Fig. 23, both R_z and R_a serve as response indices for local surface geometrical characteristics. Fig. 17(a), Fig. 22(a) and Fig. 23(a) provide evidence that R_z could serve as the criterion for detecting surface defects. Bulges and concavities on the surface could be detected by the sharp convex regions of the R_z curve. Fig. 17(b), Fig. 22(b) and Fig. 23(b) demonstrate that R_a could serve as the criterion for determining the severity of the stair stepping characteristics. The stair stepping characteristics on the surface could be quantitatively evaluated by the value of the R_a.

Table 7 displays the measurement results of S_z and S_a for Case B and Case C. From Table 3 and Table 7, S_z and S_a are response indices for global surface geometrical characteristics.S_z can serve as the criterion for assessing altitude difference (the distance from the peak to the valley on the surface). S_z increases as the altitude difference increases. By measuring the altitude differences in Case A, Case B and Case C, the altitude difference in Case A is the largest, the altitude difference in Case B is the smallest, and the altitude difference in Case C is moderate. The measured S_z for Case A, Case B and Case C are 14.0726 mm, 6.8408 mm and 7.1836 mm, respectively, which verifies the correctness of S_z as the response index for the altitude difference. Meanwhile, S_a can serve as the criterion for determining flatness level. S_a increases as the flatness level decreases. By observing the flatness levels in Case A, Case B and Case C, the flatness level in Case A is the lowest, the flatness level in Case B is the moderate, and the flatness level in Case C is highest. The measured S_a for Case A, Case B and Case C are 0.8159 mm, 0.4194 mm and 0.2396 mm, respectively, which confirms the correctness of S_a as the response index for the flatness level.

To verify the accuracy of roughness measurement, a test was conducted using a metal trapezoidal calibration block, as shown in Fig. 24. $\overline{R_z}$ and $\overline{R_a}$, S_z and S_a of the six planes were measured by two laser scans. Table 8 lists the measurement errors of roughness indices. The average

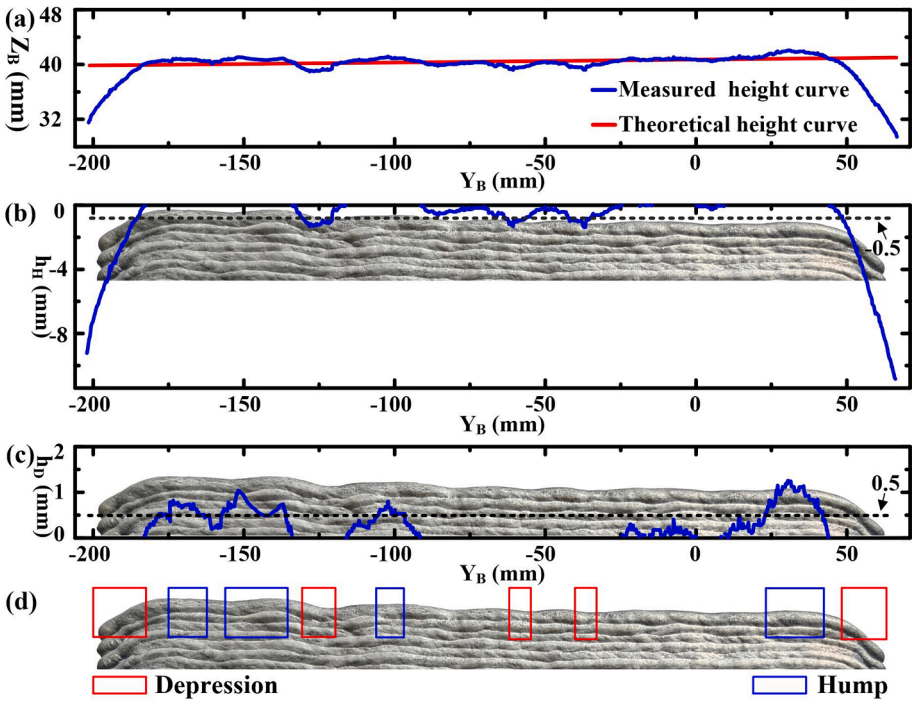

Fig. 20. Results of defect detection for Case B. (a) Measured height curve and theoretical height curve. (b) Detection results for depression. (c) Detection results for hump. (d) Positions of defects.

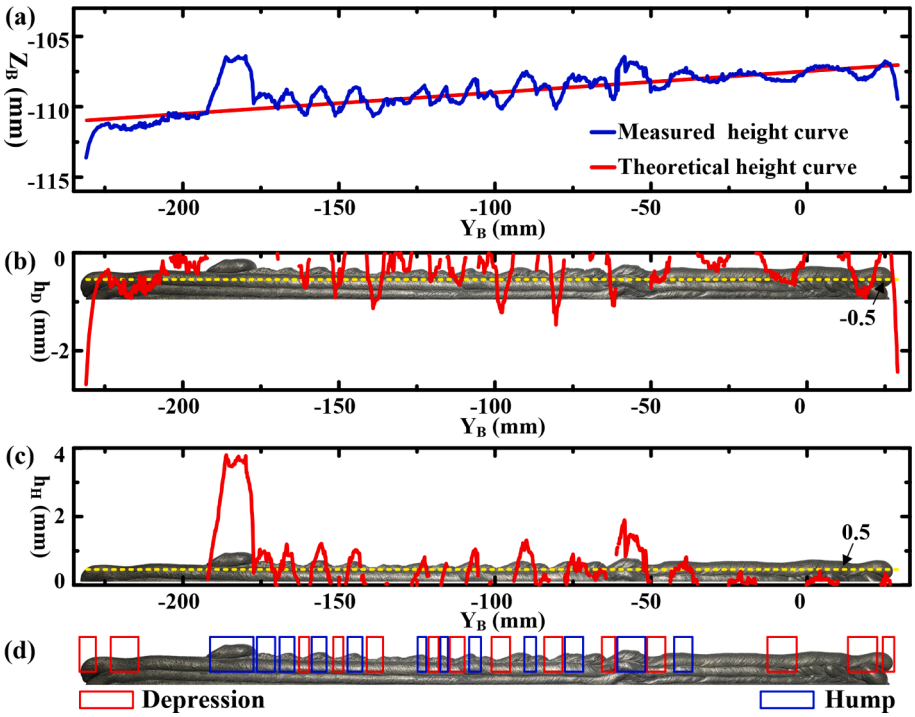

Fig. 21. Results of defect detection for Case C. (a) Measured height curve and theoretical height curve. (b) Detection results for depression. (c) Detection results for hump. (d) Positions of defects.

measurement errors for $\overline{R_z}$, $\overline{R_a}$, S_z and S_a were 0.0626 mm, 0.0176 mm, 0.1558 mm and 0.0140 mm, respectively. These measurement errors satisfy the precision requirements for industrial surface quality inspection, confirming that the roughness measurement algorithms and strategies proposed in this paper are accurate and reliable.

5.4. Performance analysis

The proposed in-situ surface inspection system establishes a scalable technological framework for online quality monitoring in the WA-DED process through synergistic integration of 3D topography reconstruction, defect detection and roughness measurement. The multi-modal data fusion strategy can be extended to real-time quality inspection in

Table 5
Defect detection results for Case A, Case B and Case C.

Real type	Algorithm detection results			Summary
	Normal	Hump	Depression	
Normal	–	0	1	1
Hump	1	16	0	17
Depression	0	0	21	21

Laser Powder Bed Fusion (LPBF), Electron Beam Deposition (EBD) and Laser Directed Energy Deposition (LDED). Notably, the system still has some limitations: The defect detection criterion based on global geometrical information may overlook local geometrical features and affect the detection accuracy; The precision of 3D topography reconstruction and roughness measurement is affected by ambient lights and interface reflections. Future studies could enhance the robustness and industrial applicability of the system by introducing local feature refinement algorithms and integrating anti-interference multi-wavelength sensing modules.

6. Conclusion

An in-situ surface inspection system for WA-DED integrating defect detection, 3D topography reconstruction and roughness measurement was developed. The following conclusions can be drawn from the work:

(1) The transformations between various coordinate systems in the laser vision inspection system are determined by the sphere-based hand-eye calibration, endowing the system with the ability to accurately perceive the 3D coordinates of spatial points, and effectively addressing the challenges related to surface point cloud recognition during the WA-DED process.
(2) The feasibility and accuracy of the proposed CIICEA and K-QCBFA algorithms in 3D topography reconstruction have been successfully demonstrated. Reconstruction experiments achieve a precision of 0.247 mm.
(3) Identification and classification of weld bead defects are realized through the established defect detection criterion by analyzing weld bead height difference curve. Experimental results indicate an accuracy of 97.44 %, with precision, recall and F1 score of 97.37 %.
(4) Quantitatively evaluations of roughness indices R_z, R_a, S_z and S_a are accomplished by analyzing vertical distances from surface points to the datum plane. Measurement errors for R_z, R_a, S_z and S_a are 0.0626 mm, 0.0176 mm, 0.1558 mm and 0.0140 mm, respectively, all of which fall within an acceptable range. Furthermore, R_z, R_a, S_z and S_a could serve as criteria for detecting surface defects, determining the severity of the stair stepping characteristic, assessing altitude difference, evaluating flatness level, respectively.

The implemented in-situ surface inspection technique can be integrated into the WA-DED production system, providing comprehensive

Table 6
Evaluation indices of algorithm performance.

Defects	TP	FP	TN	FN	Accuracy	Precision	Recall	F1
Hump	16	0	22	1	–	1	0.9412	0.9697
Depression	21	1	17	0	–	0.9545	1	0.9767
Summary	37	1	39	1	0.9744	0.9737	0.9737	0.9737

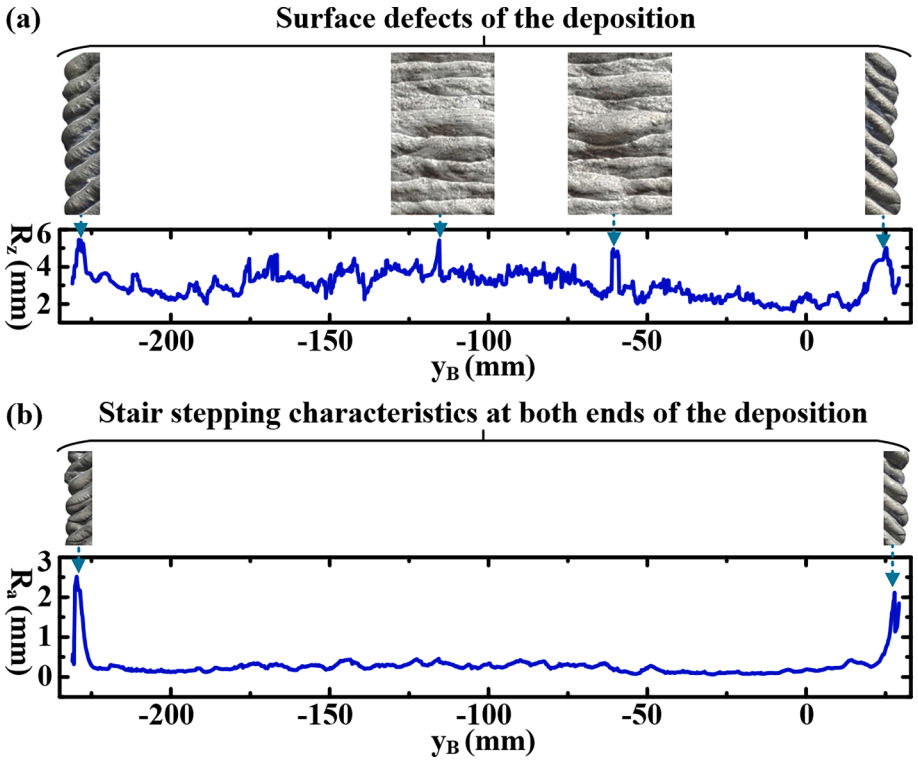

Fig. 22. Results of roughness measurement for Case B. (a) R_z. (b) R_a.

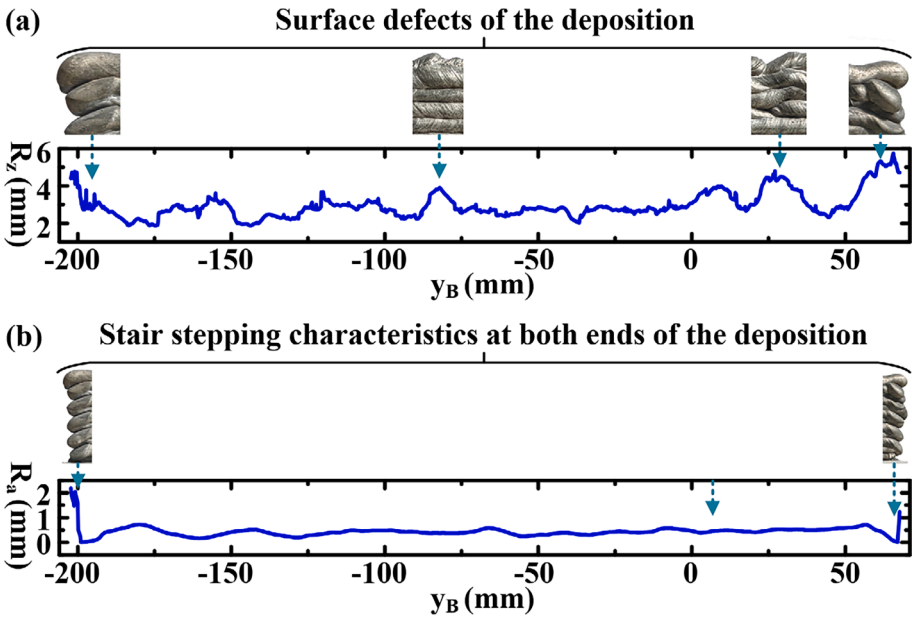

Fig. 23. Results of roughness measurement for Case C. (a) R_z. (b) R_a.

Table 7
The measurement results of S_z and S_a.

Case No.	S_z(mm)	S_a(mm)
Case A	14.0726	0.8159
Case B	6.8408	0.4194
Case C	7.1836	0.2396

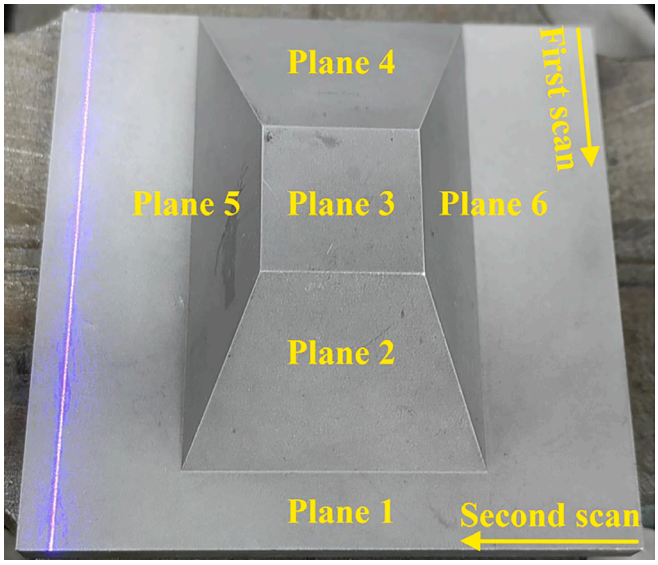

Fig. 24. Precision verification for roughness measurement.

and reliable supporting data for process improvement, quality grading, and product repair. Due to the limited laser length, the surface quality information obtained under a single scan for large components is incomplete. Future work should optimize algorithms to obtain holistic assessments via multi-scan point cloud stitching and multi-sensor fusion.

Table 8
Precision of roughness measurement.

Plane No.	$\overline{R_z}$	$\overline{R_a}$	S_z	S_a
Plane 1	0.0787	0.0248	0.1764	0.0050
Plane 2	0.0461	0.0210	0.1561	0.0131
Plane 3	0.0699	0.0180	0.1894	0.0069
Plane 4	0.1322	0.0253	0.1936	0.0254
Plane 5	0.0275	0.0091	0.1160	0.0093
Plane 6	0.0213	0.0072	0.1035	0.0242
Average error	0.0626	0.0176	0.1558	0.0140

CRediT authorship contribution statement

Hao Wang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Songyu Qi:** Validation, Formal analysis, Data curation. **Tiejun Zang:** Validation, Formal analysis, Data curation. **Chao Chen:** Methodology, Investigation, Conceptualization. **Xiaohui Zhao:** Resources, Investigation, Funding acquisition. **Yu Liu:** Resources, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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